# Mojtaba Jafarian Abyaneh

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## Summary

Ph.D. Candidate and Graduate Research Assistant at Florida Atlantic University specializing in AI and Machine Learning for transportation engineering. Builds Transformer models for Multi-Agent trajectory prediction from LiDAR, spatial graph networks for mobility flow, and city-scale CFD for hurricane pedestrian wind. Research funded by the NSF and NIH. First-author in *IEEE OJ-ITS*, with a second paper in revision for *IEEE T-ITS*.

## Education

**Florida Atlantic University** — Boca Raton, FL
*Ph.D. in Transportation and Environmental Engineering | GPA: 4.0/4.0* — *2024 – Present*
- *Research Focus: Deep learning for trajectory prediction and spatiotemporal mobility modeling*
- *Advisor: Dr. Jinwoo Jang*

**University of New Brunswick** — Fredericton, NB, Canada
*M.Sc. in Civil Engineering* — *2018 – 2020*
- *Thesis: Vibration testing of cracked reinforced concrete beams under sustained load*

**Iran University of Science and Technology** — Tehran, Iran
*M.Sc. in Civil/Earthquake Engineering* — *2014 – 2017*
- *Thesis: Detection and estimation of structural damage using finite element model updating by means of damage-sensitive dynamic parameters*

**Iran University of Science and Technology** — Tehran, Iran
*B.Sc. in Civil Engineering* — *2010 – 2014*

## Research Interests

- Deep Learning for Intelligent Transportation Systems
- Multi-Agent Trajectory Prediction
- Spatial Graph Neural Networks
- Computer Vision & LiDAR Data Processing
- Urban Transportation Safety Modeling
- Urban Digital Twins & Computational Simulation
- Spatiotemporal Flow Prediction

## Experience

**Doctoral Research Assistant** — Aug. 2024 – Present
Florida Atlantic University (I-SENSE Lab) — *Boca Raton, FL*
- Developed **SIPT** for Multi-Agent trajectory prediction: ADE 0.72 m, **30.1% better than Social LSTM** and **49% lower collision rate than a standard Transformer** (5.7% $\rightarrow$ 2.9%), on 64,000+ COSMOS trajectories; in revision for *IEEE T-ITS*.
- Developed Transformer model for real-world LiDAR-based trajectory prediction: **15.24%/13.88% ADE improvement over LSTM** (single-/Multi-Agent) on 12,390+ trajectories at a West Palm Beach intersection; published in *IEEE OJ-ITS*, Vol. 7 (2026).
- Managing the **NIH** longitudinal dataset (R01AG068472): multi-sensor data and cognitive assessments from 750 older drivers, 12 waves over 3 years.
- Co-mentored two NSF CS3 REU students: GPS-based driver behavior analysis (FMM map matching) and watertight 3D city reconstruction (QuASAR).
- Running **OpenFOAM** CFD on HPC: hurricane pedestrian wind hazard for downtown Miami.

**Graduate Research Mentor** — Jan. 2026 – Present
Vertically Integrated Projects (VIP) Program, Florida Atlantic University — *Boca Raton, FL*
- Overseeing 11 undergraduates in two VIP groups on AI smart-sensor applications for coastal resilience: computer vision, sentiment analysis, sensor analytics.

**Guest Lecturer** — Sep. 2026
Florida Atlantic University — *Boca Raton, FL*
- **CES 3102C Analysis of Structures**: taught shear force and bending moment diagrams, with my Interactive 2D Beam Solver.

**Data Analyst** Nov. 2022 – Jul. 2024
Freelance *Fredericton, NB, Canada*
- Analyzed import/export data for trade compliance, tariff classification and logistics optimization.
- Built import-volume forecasting models and Tableau dashboards for shipment performance, cost trends and supplier metrics.

**Graduate Research Assistant** Sep. 2018 – Oct. 2020
University of New Brunswick *Fredericton, NB, Canada*
- Modeled reinforced concrete beams under sustained/cyclic loading in ABAQUS (concrete damage plasticity model); conducted vibration testing with accelerometers.
- Evaluated long-term deflection, natural frequencies, and mode shapes via ACI 318-19 / CSA A23.3 effective moment of inertia formulations.

**Graduate Teaching Assistant** Jan. 2019 – Apr. 2020
University of New Brunswick *Fredericton, NB, Canada*
- **Mechanics of Materials Experimental Lab** (Jan. – Apr. 2019, Sep. – Dec. 2019): Supervised undergraduates in material testing (tensile, compression, torsion), ensuring lab safety and accurate data collection.
- **Statics Computational Lab** (Jan. – Apr. 2020): Taught computational methods and statics, including force analysis, moment calculations, and structural equilibrium concepts.

**Graduate Research Assistant** Sep. 2014 – Apr. 2017
Iran University of Science and Technology *Tehran, Iran*
- Structural health monitoring and damage detection via optimization algorithms (grey wolf, genetic) for civil infrastructure.

## Publications

1. **Jafarian Abyaneh, M.** and Jang, J. (2026) "Transformer-Based Trajectory Prediction Using LiDAR Data for Situational Awareness in Complex Urban Environments." *IEEE Open Journal of Intelligent Transportation Systems*, Vol. 7, pp. 16–28. doi:10.1109/OJITS.2025.3640002

2. Zare Hosseinzadeh, A., Ghodrati Amiri, G., **Jafarian Abyaneh, M.**, Seyed Razzaghi, S. A., and Ghadimi Hamzehkolaei, A. (2020) "Baseline updating method for structural damage identification using modal residual force and grey wolf optimization." *Engineering Optimization*, Vol. 52, No. 4, pp. 549–566. doi:10.1080/0305215X.2019.1593400

3. **Jafarian Abyaneh, M.** (2020) "Vibration testing of cracked reinforced concrete beams under sustained load." Master's thesis, University of New Brunswick.

4. Ghodrati Amiri, G., **Jafarian Abyaneh, M.**, and Zare Hosseinzadeh, A. (2016) "Model Updating-Based Approach for Damage Prognosis in Frames via Modal Residual Force." *International Journal of Civil, Environmental, Structural, Construction and Architectural Engineering*, Vol. 10, No. 8. doi:10.5281/zenodo.1125800

5. Ghodrati Amiri, G., Zare Hosseinzadeh, A., **Jafarian Abyaneh, M.** (2015) "A new two-stage method for damage identification in linear-shaped structures via Grey System Theory and optimization algorithm." *Journal of Rehabilitation in Civil Engineering*, Vol. 3, No. 2, pp. 45–58. doi:10.22075/jrce.2016.369

## Working Papers & Under Review

1. **Jafarian Abyaneh, M.**, Jang, J., Smyth, A. W., and Turkcan, M. K. (2026) "Spatial Interaction Pooling Transformer (SIPT) for Multi-Agent Trajectory Prediction in Urban Streetscapes." *IEEE Transactions on Intelligent Transportation Systems*. [In Revision]

2. **Jafarian Abyaneh, M.**, Jang, J., and Kaisar, E. I. (2025) "Spatial Graph Convolutional Network for Predicting Bike-Sharing Origin-Destination Spatiotemporal Flows." SSRN Working Paper No. 5801889. doi:10.2139/ssrn.5801889 [Preprint]

3. **Jafarian Abyaneh, M.**, Jang, J., Smith, D. K., and Lalla, J. (2026) "The Impact of Building Geometry Fidelity on Hurricane Pedestrian Wind Hazard Across Inflow Conditions." [In Preparation]

## Conference Presentations

1. **Jafarian Abyaneh, M.** and Jang, J. (Jan. 2027) "Geometry-Aware Transformer for Multi-Agent Trajectory Forecasting at Urban Intersections." *106th Transportation Research Board (TRB) Annual Meeting*, Washington, DC. Poster presentation. [Accepted]

2. **Jafarian Abyaneh, M.**, Jang, J., and Kaisar, E. I. (Jan. 2026) "Spatial Graph Convolutional Network for Predicting Bike-Sharing Origin-Destination Spatiotemporal Flows." *105th Transportation Research Board (TRB) Annual Meeting*, Washington, DC. Poster presentation.

## Workshop Presentations

1. **Jafarian Abyaneh, M.**, Jang, J., Smyth, A. W., and Turkcan, M. K. (Nov. 2026) "Neighbor-Aware Forecasting of Pedestrian and Vehicle Movement for Safer Intersections." *2026 Infrastructure Night, Department of Civil, Environmental and Geomatics Engineering, Florida Atlantic University*, Boca Raton, FL. Poster presentation.
2. **Jafarian Abyaneh, M.** and Jang, J. (May 2026) "Predicting the Crowd: Spatially-Informed Transformers for Urban Traffic Agents." *CS3 NSF Renewal Site Visit, Columbia Engineering Innovation Hub, CS3 HQ*, New York, NY. Poster presentation.
3. Smith, D. K., **Jafarian Abyaneh, M.**, and Jang, J. (May 2026) "Voxelized High-Fidelity 3D Models for City-scale Wind Simulations." *CS3 NSF Renewal Site Visit, Columbia Engineering Innovation Hub, CS3 HQ*, New York, NY. Poster presentation.
4. **Jafarian Abyaneh, M.** and Jang, J. (Mar. 2026) "Context-Aware Deep Learning for Predicting Pedestrian and Vehicle Movement Patterns in Urban Traffic Scenarios." *8th Annual Celebration of Academic Excellence, College of Engineering and Computer Science, Florida Atlantic University*, Boca Raton, FL. Poster presentation.
5. Smith, D. K., **Jafarian Abyaneh, M.**, and Jang, J. (Mar. 2026) "Testing the Accuracy of Urban 3D Models With Airflow Simulations." *Florida Undergraduate Research Conference (FURC)*, University of North Florida, Jacksonville, FL. Poster presentation.
6. **Jafarian Abyaneh, M.** and Jang, J. (Feb. 2026) "Spatial Interaction Pooling Transformer (SIPT) for Multi-Agent Trajectory Prediction." *CS3 Innovation Summit, Columbia University*, New York, NY. **Perfect Pitch Competition** (90-second research presentation) and **Poster Presentation**.
7. Smith, D. K., Lalla, J., **Jafarian Abyaneh, M.**, and Jang, J. (Feb. 2026) "Automated Voxelization Pipeline for High-Fidelity Urban Digital Twins." *CS3 Innovation Summit, Columbia University*, New York, NY. Poster presentation.

## Research Funding

- **NSF** Engineering Research Center for Smart Streetscapes (CS3), Award No. EEC-2133516 (Graduate Researcher)
- **NIH** In-Vehicle Sensors to Detect Cognitive Change in Older Drivers, Award No. R01AG068472 (Graduate Researcher)
- **FAU** Vertically Integrated Projects (VIP): Smart Sensors and AI for Coastal Destination Resilience (Graduate Research Mentor)

## Undergraduate Research Mentoring

**NSF CS3 (EEC-2133516), Digital Twins for Smart Streetscapes, FAU** Jan. 2026 – Present

- **Dana K. Smith** (Civil Eng., FAU): voxelization & CFD-ready digital twin pipelines
- **Jonathan Lalla** (CS, FAU): voxelization & CFD-ready digital twin pipelines
- **Cyrus Khan** (CS, UC Riverside; Summer REU): GPS-to-behavior-metric pipeline, Fast Map Matching (FMM)
- **Vedant Sundriyal** (CS, Lehman College; Summer REU): QuASAR, watertight 3D city reconstruction (GitHub)

**FAU VIP Program, Smart Sensors and AI for Coastal Destination Resilience** Jan. 2026 – Present

- **Environment & Sustainability:** Dana K. Smith, Iris York, Felicia Phan, James Kane, Dana Urbano Prieto, Sammie Chamberlain (FAU)
- **Safety & Security:** Jonathan Lalla, Mykyta Garkusha, Mohammed Mered, Howard Lowe, Nicolette Madrazo (FAU)

## Honors and Awards

- FAU Graduate Fellowship for Academic Excellence (Oct. 2026), $5,000 competitive award, 2026–2027
- ASCE Palm Beach Branch Scholarship (Apr. 2026), $1,000 competitive award
- New Brunswick Innovation Foundation (NBIF) Graduate Award (2018–2019)
- Esterbauer Civil Engineering Graduate Award, University of New Brunswick (2018–2020)

## Professional Affiliations

- Institute of Electrical and Electronics Engineers (IEEE), Member
- American Society of Civil Engineers (ASCE), Member

## Technical Skills

- **Programming & Tools:** Python, MATLAB, LaTeX, Typst, Git, Tableau
- **AI & Machine Learning:** PyTorch, TensorFlow, Keras
- **Data Science:** Pandas, NumPy, Scikit-learn, GeoPandas, ArcGIS
- **Engineering Software:** ABAQUS, OpenFOAM (HPC), ParaView, SAP2000, ETABS, SAFE, Synchro, Civil 3D, MicroStation, AutoCAD